ANALYTICS ASSIGNMENT · CASE STUDY

Transaction Data Analysis

SQL Queries & Statistical Analysis of Payment Transactions · Jan – Jul 2020

DATASET
501 Transactions · 10 Users

CHANNELS
IMPS · IFT · UPI · NEFT · RTGS

DATABASE
SQLite (analysis_queries.sql)

QUESTIONS
5 Problems Solved

0 Dataset Overview

The dataset was provided as a table inside a PDF. It was extracted using PyMuPDF, verified for duplicates and missing values (none found), and loaded into a SQLite table called `transactions`. All monetary values are in Indian Rupees (₹).

501	10	7
TOTAL TRANSACTIONS	UNIQUE USERS	MONTHS (JAN–JUL 2020)

352	149	5
CREDIT TRANSACTIONS	DEBIT TRANSACTIONS	PAYMENT CHANNELS

Schema

Column	Type	Description
Transaction_Time	TEXT (date)	Date of transaction
User_Id	INTEGER	User identifier (1–10)
Transaction_Amt	REAL	Amount in ₹
Narration	TEXT	Payment channel: IMPS, IFT, UPI, NEFT, RTGS
Transaction_Type	TEXT	CREDIT or DEBIT
Txn_id	TEXT	Unique transaction identifier
Month	INTEGER	Month number (1–7)

Channel Breakdown (All Transactions)

Channel	Count	Share	In Scope (Brief)
IMPS	272	54.3%	✅ Yes
IFT	89	17.8%	❌ Not listed
UPI	86	17.2%	✅ Yes
NEFT	33	6.6%	✅ Yes
RTGS	21	4.2%	✅ Yes

1 7th Highest Debit Amount Through IMPS

SQL Query

```sql
-- Filter IMPS DEBIT transactions, sort descending, pick row 7.
-- OFFSET 6 skips 6 rows → the 7th row. No ties at this rank.

SELECT  Transaction_Amt
FROM    transactions
WHERE   Narration        = 'IMPS'
  AND   Transaction_Type = 'DEBIT'
ORDER BY Transaction_Amt DESC
LIMIT  1 OFFSET 6;
```

Results — Top 10 IMPS Debit Amounts

Rank	Amount (₹)	User	Date	Txn ID
1	₹9,987	9	08-Apr-2020	613AFBB8
2	₹9,887	10	09-Jul-2020	EE06FC09
3	₹9,872	3	24-Feb-2020	7641EB77
4	₹9,863	5	14-Jul-2020	834C909F
5	₹9,743	4	16-May-2020	53258869
6	₹9,566	3	28-May-2020	8D2D3F20
★ 7	**₹9,525**	**8**	**16-Jan-2020**	**64B08A24**
8	₹9,450	1	28-Jan-2020	BDF20BCA
9	₹9,434	7	17-Apr-2020	B0EEB2E7
10	₹9,337	4	11-Jun-2020	47625EBE

✅ **Answer:** The 7th highest IMPS DEBIT amount is **₹9,525** (User 8, 16-Jan-2020, Txn 64B08A24). There are 97 IMPS debit transactions in total; no ties exist at this rank.

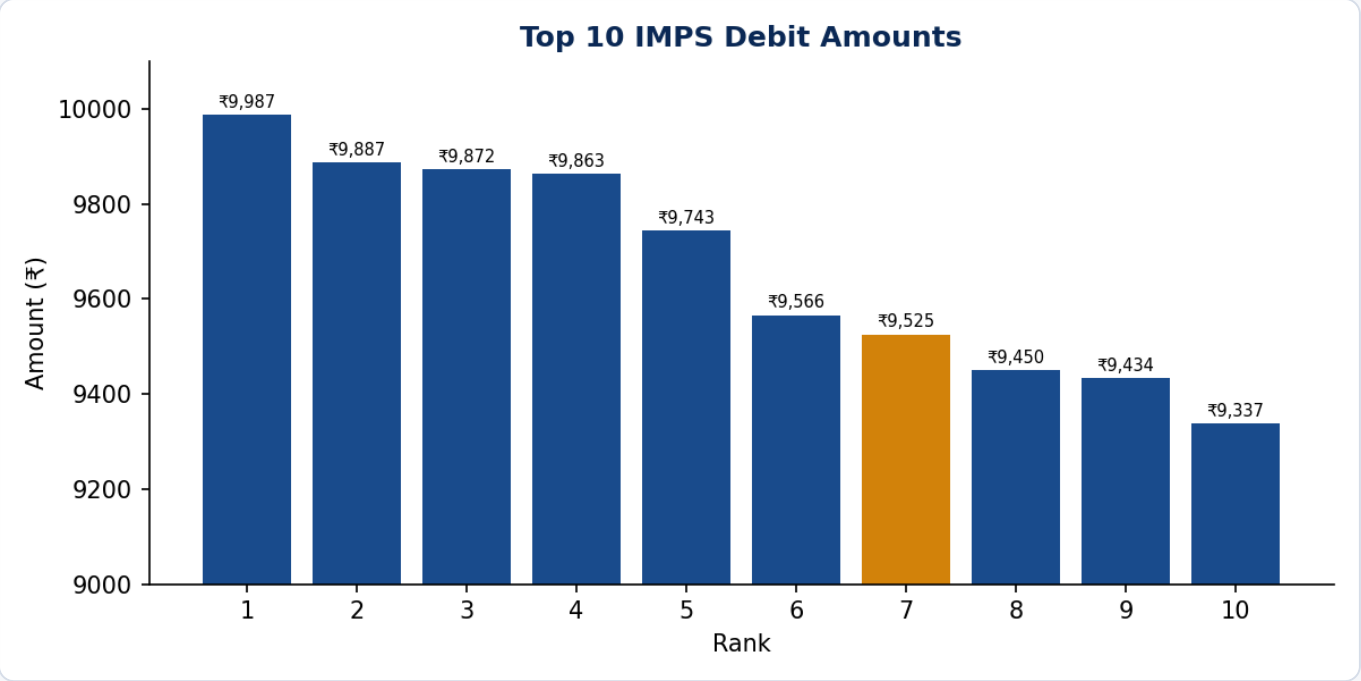


Fig 1. Top 10 IMPS DEBIT amounts — 7th rank highlighted in amber (₹9,525)

2 Number of Transactions, Category-Wise

> **Note on IFT:** The dataset contains 89 IFT (Internal Funds Transfer) transactions. Since the assignment brief specifies UPI, IMPS, RTGS, and NEFT only, IFT is excluded from this answer but is included in the channel overview table above.

SQL Query

```sql
SELECT  Narration           AS Category,
        COUNT(*)            AS No_Of_Transactions
FROM    transactions
WHERE   Narration IN ('UPI', 'IMPS', 'RTGS', 'NEFT')
GROUP BY Narration
ORDER BY No_Of_Transactions DESC;
```

Results

Category	Transactions	Share of in-scope
IMPS	272	65.4%
UPI	86	20.7%
NEFT	33	7.9%
RTGS	21	5.0%
★ **Total (in scope)**	**412**	**100%**

✅ **Answer:** IMPS: **272** | UPI: **86** | NEFT: **33** | RTGS: **21**

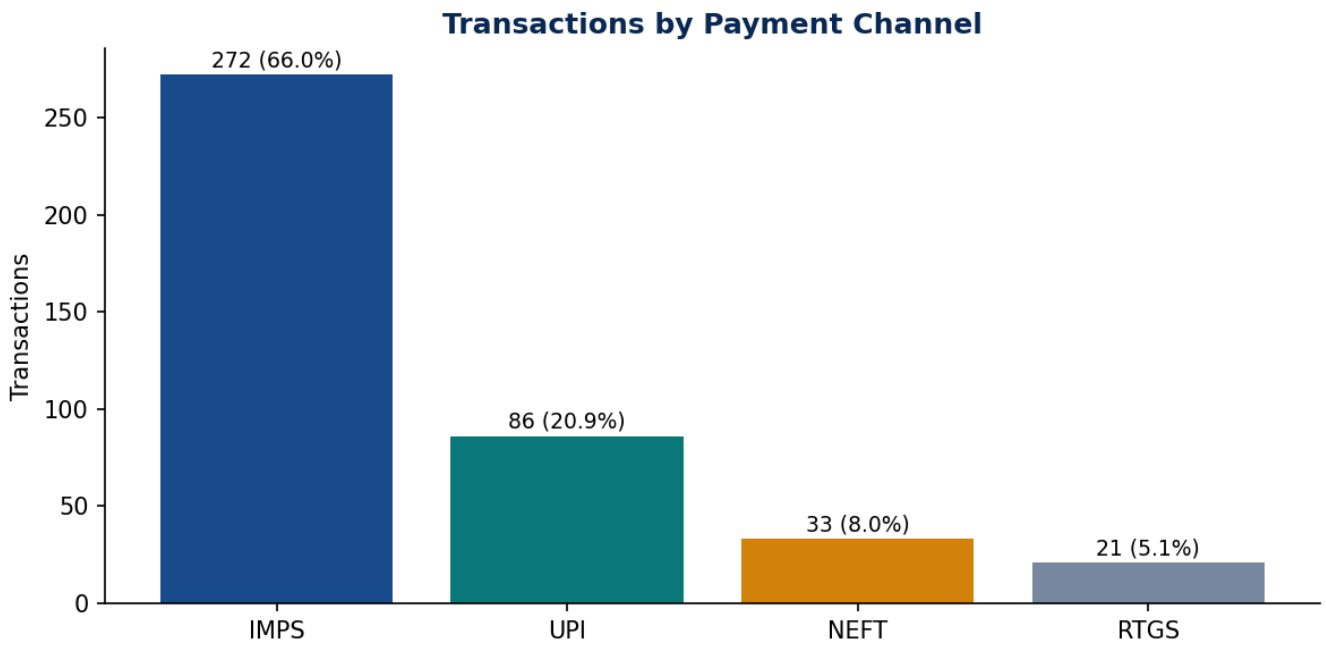


Fig 2. Transaction count by payment channel. IMPS dominates at 65% of in-scope transactions.

INSIGHT

IMPS accounts for nearly two-thirds of all in-scope volume, reflecting its popularity for immediate peer-to-peer and merchant transfers. UPI follows at ~21%. RTGS, typically used for high-value inter-bank transfers, has the lowest count, which is expected given the small-to-mid range of amounts in this dataset (₹108–₹9,987).

3 Bell Curve & Box Plot — Load Amount

Assumption: The dataset has no column named "load amount". This is interpreted as the amount of **CREDIT transactions** — money credited (loaded) into the account — covering 352 transactions. This is the most consistent interpretation given standard fintech terminology.

Statistical Summary

352 COUNT	**₹4,894** MEAN	**₹4,708** MEDIAN	**₹2,914** STD DEV
₹108 MIN	**₹9,950** MAX	**₹2,329** Q1 (25TH PCT)	**₹7,492** Q3 (75TH PCT)

₹5,163	0.049	−1.239	0
IQR	SKEWNESS	KURTOSIS	OUTLIERS DETECTED

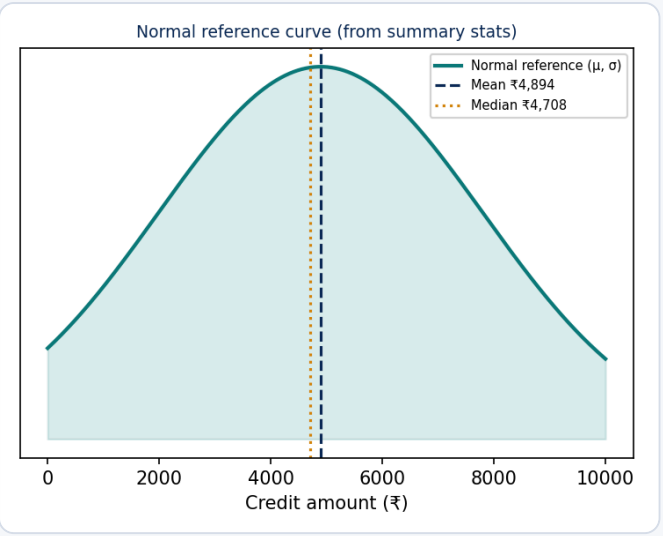


Fig 3a. Normal reference curve (μ = ₹4,894, σ = ₹2,914)

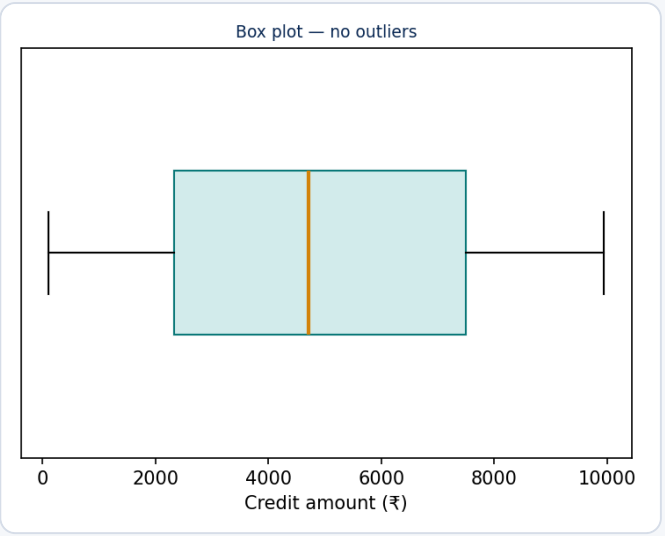


Fig 3b. Box plot with IQR and outlier indicators

STATISTICAL INTERPRETATION

Symmetry: Mean (₹4,894) and Median (₹4,708) are close, and skewness of **0.049** (near-zero) confirms the distribution is approximately symmetric — neither left- nor right-skewed. The bell curve closely mirrors the reference normal distribution.

Platykurtic shape: Kurtosis of **−1.239** indicates the distribution is flatter than a normal curve (platykurtic). Amounts are spread fairly evenly across the ₹108–₹9,950 range rather than clustering near the mean — suggesting users credit amounts across all sizes rather than favouring a particular band.

No outliers: The IQR fences (Q1 − 1.5×IQR = −4,416 → floored at ₹108; Q3 + 1.5×IQR = ₹15,237) contain all observations. No extreme outlier transactions exist.

4 Monthly Cohort View of Active Users (DEBIT)

A user's **cohort** is the month of their first DEBIT transaction. Each cell shows how many distinct users from that cohort made at least one DEBIT transaction in the given activity month.

SQL Query

```sql
WITH debit_txns AS (
    SELECT User_Id,
           strftime('%Y-%m', Transaction_Time) AS Txn_Month
    FROM   transactions
    WHERE  Transaction_Type = 'DEBIT'
),
cohort AS (
    SELECT User_Id, MIN(Txn_Month) AS Cohort_Month
    FROM   debit_txns GROUP BY User_Id
),
activity AS (
    SELECT DISTINCT d.User_Id, d.Txn_Month, c.Cohort_Month
    FROM   debit_txns d JOIN cohort c ON d.User_Id = c.User_Id
)
SELECT Cohort_Month,
       Txn_Month        AS Active_Month,
       COUNT(DISTINCT User_Id) AS Active_Users
FROM   activity
GROUP BY Cohort_Month, Txn_Month
ORDER BY Cohort_Month, Txn_Month;
```

Cohort Matrix

Cohort / Active Month	2020-01	2020-02	2020-03	2020-04	2020-05	2020-06	2020-07
2020-01	8	5	8	7	7	7	8
2020-02	—	1	—	1	1	—	1
2020-03	—	—	1	1	1	1	1
2020-04	—	—	—	—	—	—	—
2020-05	—	—	—	—	—	—	—
2020-06	—	—	—	—	—	—	—
2020-07	—	—	—	—	—	—	—

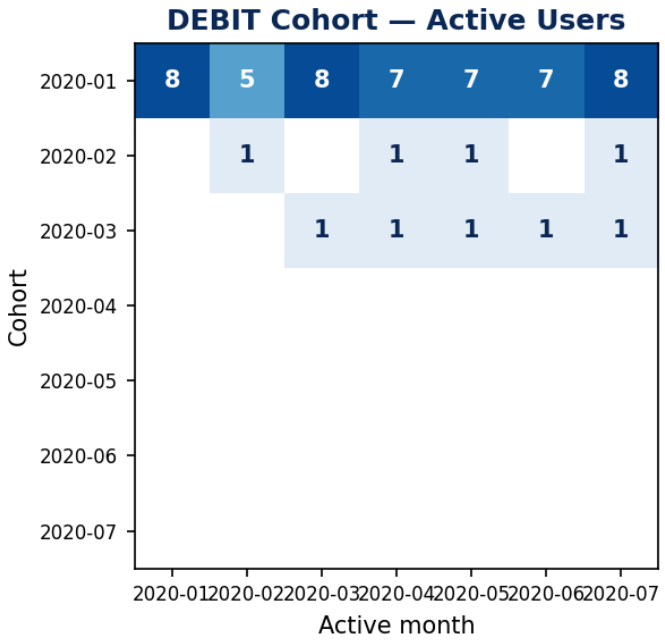


Fig 4. Monthly cohort heatmap — darker cells indicate higher active user counts

INTERPRETATION

8 of the 10 users made their first DEBIT transaction in January 2020. One additional user joined in February, and the last in March — after March, no new cohorts emerge (April–July cohort rows are all blank).

The **Jan-2020 cohort remains consistently active** throughout the 7-month period (5–8 users active each month), with no clear retention drop-off. This suggests a stable, recurring user base rather than one-time transactors. The Feb and Mar cohorts each contribute 1 user, staying active intermittently.

5 Top 10 Percentile Users by Net Amount (DEBIT − CREDIT)

Net Amount = Total DEBIT − Total CREDIT for each user. The top 10 percentile means `PERCENT_RANK() ≥ 0.90` . With only 10 users, this isolates the single user with the least-negative (highest) net amount.

SQL Query

```sql
WITH user_net AS (
    SELECT User_Id,
           SUM(CASE WHEN Transaction_Type = 'DEBIT'
                    THEN  Transaction_Amt ELSE 0 END) AS Total_Debit,
           SUM(CASE WHEN Transaction_Type = 'CREDIT'
                    THEN  Transaction_Amt ELSE 0 END) AS Total_Credit,
           SUM(CASE WHEN Transaction_Type = 'DEBIT'
                    THEN  Transaction_Amt
                    ELSE -Transaction_Amt END)   AS Net_Amount
    FROM   transactions GROUP BY User_Id
),
ranked AS (
    SELECT User_Id, Total_Debit, Total_Credit, Net_Amount,
           PERCENT_RANK() OVER (ORDER BY Net_Amount) AS Pct_Rank
    FROM   user_net
)
SELECT User_Id, Total_Debit, Total_Credit, Net_Amount
FROM   ranked
WHERE  Pct_Rank >= 0.90
ORDER BY Net_Amount DESC;
```

Full Ranked List — All 10 Users

Rank	User	Total Debit (₹)	Total Credit (₹)	Net Amount (₹)
★ 1 (Top 10%)	8	₹1,29,734	₹1,65,538	−₹35,804
2	1	₹73,830	₹1,11,143	−₹37,313
3	3	₹1,06,386	₹1,46,026	−₹39,640
4	10	₹94,295	₹1,52,722	−₹58,427
5	4	₹1,00,477	₹1,88,518	−₹88,041
6	7	₹75,500	₹1,88,696	−₹1,13,196
7	6	₹59,334	₹1,76,723	−₹1,17,389
8	9	₹58,316	₹1,86,106	−₹1,27,790
9	2	₹70,901	₹2,03,354	−₹1,32,453
10	5	₹56,298	₹2,03,953	−₹1,47,655

✅ **Answer: User 8** is the top 10 percentile user with a net amount of **−₹35,804** (smallest net outflow). Every user has a negative net amount across this dataset, meaning credits consistently exceed debits — User 8 has the smallest gap.

OBSERVATION

All 10 users show negative net amounts (Total DEBIT − Total CREDIT < 0), meaning this dataset predominantly reflects credit/inflow activity. The "top 10 percentile by net amount" therefore identifies the user whose spending (DEBIT) is closest to their income (CREDIT). User 8 has the most balanced debit-to-credit ratio among all users.

6 Limitations & Assumptions

- ! **"Load amount" assumption:** Interpreted as the amount of CREDIT transactions. The dataset has no explicit "load amount" column. This assumption is stated clearly throughout Q3.
- ! **IFT channel:** 89 IFT transactions exist in the dataset but are not included in Q2 results, as the brief specifies only UPI, IMPS, RTGS, NEFT.
- ! **Small user base:** With only 10 users, the top 10 percentile collapses to a single user (Q5). Statistical distributions and cohort patterns should be interpreted with this small sample size in mind.
- ! **Date parsing:** Dates are in M/D/YYYY format; SQLite's `strftime()` requires ISO format. In Q4, the Month column (integer 1–7) was used directly for grouping to avoid date-parsing edge cases.
- ! **No deduplication needed:** Txn_id values were verified to be unique across all 501 records.
- ! **Currency:** All amounts are assumed to be in Indian Rupees (₹). No currency column exists in the dataset.